

SAI RUCHITH REPALA

+1 316-518-5677 | ruchithkumar0107@gmail.com

PROFESSIONAL SUMMARY:

Dynamic **PySpark Developer** with **5 years** of IT experience in software development, big data technologies, and analytical solutions. Proficient in designing and implementing scalable data pipelines, ETL workflows, and machine learning models using **PySpark**, **Hadoop**, and cloud platforms like **Azure and AWS**. Proficient in Spark Core, Spark SQL, and Spark Streaming for large-scale data processing. Experienced in working with Hadoop ecosystem tools such as HDFS, Hive, HBase, Sqoop, and Oozie, as well as NoSQL databases like Cassandra and MongoDB. Skilled in data modeling, performance optimization, and integrating Spark applications with cloud storage solutions. Strong expertise in Python, Scala, and SQL, leveraging libraries like Pandas, NumPy, and Scikit-learn for data transformation and predictive analytics. Collaborative and detail-oriented, with a proven ability to build and deploy industrial-scale data lakes, optimize distributed data processing, and drive business insights through advanced data engineering solutions.

EDUCATION:

Master of Science in Computer Science | *Wichita State University, Wichita, KS*

Bachelor of Technology in Computer Science Engineering | *GITAM University, Visakhapatnam, India*

TECHNICAL SKILLS:

- **Programming & Databases:** Python, SQL, Java, R, MySQL, MS SQL Server, Oracle, Teradata, PL/SQL
- **Big Data Technologies:** PySpark, Apache Spark (Core, SQL, Streaming, MLlib), Hadoop Ecosystem (HDFS, MapReduce, YARN, Hive, Pig, HBase, Sqoop, Oozie, Zookeeper, Flume, Impala)
- **Cloud & Data Platforms:** AWS (EC2, S3, EMR), Azure (Databricks, Data Lake), Google Cloud (BigQuery)
- **Data Processing & ETL:** Apache Airflow, Informatica, Talend, Data Modeling, Data Warehousing, Dimensional Modeling
- **Visualization & BI Tools:** Tableau, Power BI, QlikView, MicroStrategy, Pentaho
- **Python Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Plotly, SciPy, PyTorch
- **NoSQL Databases:** MongoDB, Cassandra, HBase
- **Methodologies & Collaboration:** Agile/Scrum, SDLC, Git/GitHub, Jira, Confluence
- **Scheduling & Workflow Automation:** Apache NiFi, UC4, Cisco Tidal, Autosys
- **Testing & Performance Optimization:** JUnit, MRUnit, Performance Tuning (Spark, Hive), Kerberos Authentication

PROFESSIONAL EXPERIENCE:

Freddie Mac

Senior PySpark Developer | McLean, VA | Feb 2023 - Till now

- Built and optimized ETL pipelines using Apache Spark, Hive, and HDFS, ensuring smooth data ingestion and transformation.
- Integrated PySpark with AWS S3 and Azure Databricks, improving cloud-based data processing capabilities.
- Implemented data validation and cleansing strategies using PySpark, reducing data inconsistencies by 30%.
- Designed and deployed real-time streaming solutions using Spark Streaming and Kafka, enabling faster financial risk analysis.
- Conducted data modeling and performance tuning for Spark applications, improving query performance by 50%.
- Integrated AWS X-Ray SDK to trace PySpark data pipelines, providing full visibility into distributed processing flows and enhancing performance monitoring.
- Utilized Artifactory for managing and distributing PySpark job dependencies and custom Python packages, ensuring environment consistency across staging and production.
- Added AWS X-Ray annotations and metadata in Spark jobs for granular performance metrics, aiding in end-to-end traceability.
- Streamlined artifact versioning for Spark JARs and Python wheels through automated publishing to Artifactory as part of CI/CD process.
- Developed robust data ingestion and transformation scripts using Python, optimizing workflows for financial risk modeling and data analysis.

- Developed custom UDFs and Spark SQL transformations to process structured and semi-structured data from multiple sources. Collaborated with data analysts, engineers, and business stakeholders to translate financial requirements into scalable data solutions.
- Automated data workflows using Apache Airflow and Oozie, ensuring timely execution of batch processes.
- Conducted code optimization and debugging to improve processing speeds and reduce memory overhead.
- Ensured compliance with financial data governance by implementing role-based access controls and encryption for sensitive datasets.
- Led unit testing and integration testing using PyTest and JUnit, ensuring data accuracy and reliability across pipelines.
- Developed interactive data visualization dashboards using Tableau and Power BI, providing insights for mortgage and credit risk assessments.

Deloitte

PySpark Developer / Bengaluru, India / Oct 2020 - Nov 2022

- Developed and optimized PySpark scripts to process large datasets efficiently in a distributed computing environment.
- Implemented ETL pipelines using PySpark and Hadoop ecosystem tools like Hive, HDFS, and Sqoop for structured and semi-structured data.
- Designed and built data ingestion workflows to move data from various sources to HDFS, HBase, and Hive for downstream analytics.
- Created Spark DataFrames and RDDs to transform and aggregate large-scale datasets, improving processing efficiency.
- Developed real-time data processing pipelines using Spark Streaming to handle continuous data flows.
- Integrated PySpark with Hadoop YARN for resource-efficient cluster computing.
- Optimized Spark performance by implementing partitioning, caching, and broadcasting to enhance query execution speed.
- Used Hive and Spark SQL to perform advanced analytical queries and data transformations.
- Developed and maintained Python-based automation scripts for scheduling and monitoring data workflows.
- Implemented job orchestration using Oozie, Apache Airflow, and UC4 to automate and manage ETL jobs.
- Worked with Azure Data Services, including Azure Data Lake, Azure Synapse, and Azure Databricks, to process large datasets in the cloud. Developed data ingestion and transformation pipelines in PySpark, integrating with Artifactory to manage reusable modules and internal Python libraries.
- Assisted DevOps teams in configuring Artifactory repositories for efficient dependency management in Spark and Hadoop ecosystems.
- Implemented AWS X-Ray instrumentation to monitor PySpark data flows on AWS-based services, providing operational insights into ETL performance.
- Automated testing and deployment of PySpark applications with CI/CD pipelines, leveraging Artifactory for artifact lifecycle management.
- Automated data workflows using Python to manage data ingestion, cleansing, and reporting tasks across cloud-based data lakes.
- Designed modular Python utilities for ingestion, transformation, and unit testing, promoting clean code practices and reusability.
- Designed and implemented data lake architectures for efficient storage, retrieval, and processing of structured and unstructured data.
- Utilized Kafka and Apache Flume to ingest real-time streaming data into Hadoop-based data lakes.
- Collaborated with data scientists and analysts to provide pre-processed datasets for machine learning and analytical use cases.
- Implemented data validation and quality checks using PySpark UDFs and Python-based unit testing frameworks.

Zerodha

PySpark Developer | Bengaluru, India | Sep 2019 - September 2020

- Developed and optimized PySpark scripts for processing large-scale financial data to enhance data pipeline efficiency.
- Designed and implemented ETL workflows using Spark SQL, DataFrames, and RDDs to support high-volume data ingestion from multiple sources.
- Built real-time data streaming solutions using Apache Kafka and Spark Streaming to reduce data processing latency.
- Integrated Hive and HBase with PySpark for seamless querying and analytics on structured and semi-structured financial datasets.
- Developed and deployed machine learning pipelines in PySpark for fraud detection and risk assessment in stock trading.
- Utilized Azure Data Lake and Databricks to manage and analyze large-scale trading and market data efficiently.
- Performed data cleansing, transformation, and aggregation using PySpark UDFs and window functions to generate insights for market trends.
- Collaborated with data engineers and analysts to improve data quality, schema design, and performance tuning of PySpark jobs.
- Implemented Delta Lake for versioning and ACID transactions on trading data, ensuring data consistency across systems.
- Automated and scheduled PySpark jobs using Apache Airflow to minimize manual interventions.
- Enabled trace-level debugging of Kafka–Spark Streaming pipelines using AWS X-Ray, improving transparency across real-time jobs.
- Managed Python packages and Spark dependencies using Artifactory, ensuring version consistency and reducing deployment issues.
- Implemented metadata tagging in X-Ray traces for Spark Streaming jobs to capture processing times and Kafka lag metrics.
- Built data processing utilities using Python to analyze trade behavior, detect anomalies, and generate financial reports.

PROJECTS:

1. Python Project: Fraud Detection System:

- Built a real-time fraud detection pipeline using PySpark, Kafka, and Spark Streaming, reducing false positives by 18%.
- Developed Random Forest and Time-Series models with Pandas, NumPy, and Scikit-learn to detect transactional anomalies.

1. SQL Project: Customer Churn Analysis

- Designed SQL-based churn prediction models using Window Functions, CTEs, and Joins, improving retention by 12%.
- Automated churn reports and stored procedures, enabling real-time customer behavior tracking.

2. Azure Project: Retail Sales Insights & Data Pipeline

- Built a cloud-based data pipeline using Azure Data Lake, Synapse Analytics, and Data Factory to process sales data.
- Created Power BI dashboards for regional sales insights, driving a 10% increase in sale